

Skills Summary

Geometric Series, p -Series, and the n th-Term Test

Use this reference while completing your worksheet or when studying.

Skill 1 — Geometric series: classify by r , then sum

Shape: $\sum ar^k$ — every term is a fixed multiple r of the one before it. Find r by dividing any term by its predecessor.

Test: converges exactly when $|r| < 1$; diverges when $|r| \geq 1$. The sign of r only makes the terms alternate.

Sum (only when $|r| < 1$): $S = \frac{a}{1-r}$, where a is the *first term actually written*, not the coefficient. Substitute the starting index to find it.

Example: Find $\sum_{n=0}^{\infty} 5 \left(\frac{1}{3}\right)^n$.

Step 1. Consecutive terms differ by the fixed factor $\frac{1}{3}$, so this is geometric and the sum formula applies once $|r| < 1$ is confirmed.

Step 2. $a = 5 \left(\frac{1}{3}\right)^0 = 5$ and $r = \frac{1}{3}$, so $S = \frac{5}{1 - \frac{1}{3}} = \frac{5}{\frac{2}{3}}$
 $\Rightarrow S = 15/2$

Try it: Find $\sum_{n=1}^{\infty} 8 \left(\frac{1}{2}\right)^{n-1}$.

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Skill 2 — p -series: classify by p alone

Shape: $\sum \frac{1}{n^p}$ with $p > 0$ constant. Rewrite roots as powers first: $\sqrt{n} = n^{1/2}$, $\sqrt[3]{n^2} = n^{2/3}$.

Test: converges when $p > 1$; diverges when $p \leq 1$. That single comparison is the whole test.

The borderline case is $p = 1$, the harmonic series $\sum \frac{1}{n}$, and it *diverges*. The test gives no sum — only a verdict.

Example: Classify $\sum_{n=1}^{\infty} \frac{1}{n^{4/3}}$.

Step 1. The terms are a constant power of n in the denominator, so this is a p -series and the only thing to decide is where p sits relative to 1.

Step 2. Here $p = \frac{4}{3}$, and $\frac{4}{3} = 1.333\dots$ is greater than 1

$\Rightarrow 4/3 > 1$, so the series converges

Try it: Classify $\sum_{n=1}^{\infty} \frac{1}{n^{2/5}}$ — write the comparison of p with 1.

check: $\lim a_n = 0$, $2/5 < 1$, diverges

Skill 3 — The n th-term test: a divergence detector only

Statement: if $\lim_{n \rightarrow \infty} a_n \neq 0$ (or the limit fails to exist), then $\sum a_n$ **diverges**.

What it never does: a limit of 0 proves nothing at all. Say “inconclusive” and reach for another test.

Doing the limit: for a ratio of polynomials, compare degrees — equal degrees give the ratio of leading coefficients (nonzero, so the series diverges), and a bigger denominator degree gives 0 (inconclusive).

Example: Apply the n th-term test to $\sum_{n=1}^{\infty} \frac{4n+3}{n+7}$.

Step 1. Nothing here is geometric or a p -series, so start with the cheapest test — look at what the terms themselves approach.

Step 2. Numerator and denominator are both degree 1, so divide by n : $\lim_{n \rightarrow \infty} \frac{4 + 3/n}{1 + 7/n} = \frac{4}{1}$

$\Rightarrow \lim a_n = 4 \neq 0$, so the series diverges

Try it: Apply the n th-term test to $\sum_{n=1}^{\infty} \frac{5n^2 - 2}{2n^2 + n}$ — find $\lim a_n$.

check: $\lim a_n = 5/2$, diverges

(!) Watch out: “the terms go to zero, so it converges” is the single most expensive error in this unit. It is the n th-term test read backwards. Both $\sum \frac{1}{n}$ (divergent) and $\sum \left(\frac{1}{2}\right)^n$ (convergent) have terms going to zero — the limit alone cannot tell them apart.